Fiscal Year:	FY 2026	**Task Last Updated:**	FY 03/26/2026
PI Name:	Hammond, Timothy M.D.		
Project Title:	Protection from Cosmic Radiation		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	timothy.hammond@duke.edu	**Fax:**	FY 9193238644
PI Organization Type:	NON-PROFIT	**Phone:**	9194285055
Organization Name:	Institute of Medical Research/Department of Veterans Affairs		
PI Address 1:	Institute For Medical Research, Inc.		
PI Address 2:	508 Fulton St, Building 15, Room 109		
PI Web Page:			
City:	Durham	**State:**	NC
Zip Code:	27705-3875	**Congressional District:**	1
Comments:	Alternate email: Timothy.Hammond@va.gov NOTE: PI previously at Tulane University until 2007 (Ed., 09/2010) Joint appointments Professor, Department of Internal Medicine, Nephrology Division, Duke University School of Medicine, and Staff Physician, Durham VA Health Care System.		
Project Type:	Ground	**Solicitation / Funding Source:**	2022 Space Biology NNH22ZDA001N-SBBLEO E.11: Research Pathfinder
Start Date:	06/01/2024	**End Date:**	05/31/2027
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Jones, Harry	**Contact Phone:**	650.604.5518
Contact Email:	harry.jones@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Birdsall, Holly Ph.D. (Institute For Medical Research, Inc.) Panikkanvalappil, Sajanlal Ph.D. (Dana-Farber Cancer Institute, Inc) Smolikove, Sarit Ph.D. (University of Iowa, Iowa City) Wells, Bill M.B.A. (The Bionetics Corporation)		
Grant/Contract No.:	80NSSC24K1019		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Significant DNA damage from ionizing radiation is a major medical challenge for human exploration of the solar system. The Artemis missions emphasize the importance of exploring the effects of simulated deep space radiation. The nematode C. elegans is an ideal biologic model for these studies for many reasons. C. elegans has forgiving and simple culture requirements; it has shown robust growth in previous space flight experiments; the entire genome has been sequenced; and tools for genomic manipulation are freely available. Furthermore, the molecular machinery for repair of radiation damage in C. elegans is well defined. DNA damage leads to a wave of the repair proteins RPA-1, followed by RAD-51, that repair double stranded breaks. One can generate a biologic detector of DNA damage by deleting the RPA-1 gene so that the DNA damage accumulates without intervening repair. Hence, our first aim is to study the RPA-1 and RAD-51 radiation repair system in C. elegans after exposure to cosmic radiation. Histone gamma-H2AX has been a popular molecule to monitor and quantitate radiation induced foci following radiation damage. However, C. elegans lacks gamma-H2AX or a homolog. Homologous recombination is not the only metabolic pathway important in radiation protection. The broad family of intrinsically disordered proteins (IDPs) have several potential roles, termed molecular shield functions, including protection from radiation. Like tardigrades, rotifers, and some arthropods, nematodes express molecular shield proteins. The C. elegans late embryogenesis abundant (LEA) proteins are a biologically important, intrinsically disordered protein molecular shield whose active protein motif is a single 44 amino acid sequence. LEA proteins inserted into E.coli protect from radiation damage, showing LEA is sufficient to induce radiation protection. Our second aim is to study whether LEA can protect C. elegans DNA from damage by simulated cosmic radiation. Newer methods offer unprecedented resolution to the identification and quantification of radiation damage. Surface-enhanced Raman spectroscopy (SERS) is a non-destructive chemical analysis technique that can detect and identify diverse forms of DNA damage. Raman spectroscopy can be performed in situ in intact tissues, or on isolated DNA samples. Options for ground-based controls have improved dramatically. Recent improvements in laser handling equipment at the NASA Space Radiation Laboratory (NSRL) at Brookhaven National Laboratory have resulted in a simple and practical five-beam simulation of cosmic radiation. Ground-based studies are more accessible and relatively inexpensive compared to flight studies. Ground controls can include rotating suspension conditions to simulate effects of microgravity, which are known to attenuate the DNA repair process. Hypothesis and Specific Aims Hypothesis: The C. elegans intrinsically disordered protein molecular shield, late embryogenesis abundant (LEA) protein, provides protection from simulated cosmic radiation damage. Specific Aim 1: To characterize simulated cosmic radiation damage to C. elegans DNA. Specific Aim 2: To test whether the intrinsically disordered protein molecular shield, late embryogenesis abundant (LEA) protein, protects from simulated cosmic radiation damage.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The two complementary approaches we are using to characterize DNA damage will increase our understanding of the biologic responses and impacts of deep space environment on the physiology of an actively growing multicellular organism as a model of human physiology. LEA, as a molecular shield that protects against radiation damage, can lead to the development of biologics, including gene therapies that can be used in diverse organisms. Scientific return will be expanded by adding these samples to the NASA GeneLab Data System and the NASA Ames Life Sciences Data Archive for access by the research community. These studies will advance NASA technical and operational experience and lessons learned for the design and implementation of experiments using animal models for future Artemis platforms. This grant represents an immense opportunity to complete development of biological protection from cosmic radiation, central to the Moon and Mars initiatives. This work is based on the only published data on biological samples exposed to true cosmic radiation on Artemis I (Sci Rep: 2025 Oct 29;15(1):37734) including identification of Dsup as a biological protection modality.
	PROGRESS Specific Aim 1A -- Enumeration of radiation-induced foci of green fluorescent protein (GFP)-linked Replication Protein A-1 (RPA-1). Bottom line up front (BLUF): A major worm culture bottleneck has been resolved by CRISPR-editing the sC1 balancer (let-805 deletion), validation of micro irradiation methods on the new Nikon/Yokogawa super-resolution microscopy SoRa system, and the establishment of strains and crosses (including histone markers) necessary to reliably enumerate radiation-induced GFP::RPA-1 foci. Specific details: In specific aim 1, the strain for the experiment is generated: fem-2(e2105)/sC1(s2023) [dpy-1(s2170) umnIs21] III ; rad-51(iow53[GFP::rad-51])/nT1 [umnIs49] (IV, V). The new progress includes: that we can grow this strain in spinpods, a form of suspension culture to simulate microgravity, as promised in the grant, and have the protocol set. However, the strain throws a lot of sC1 homozygotes that are unusable since they are dpy (dumpy/short) phenotype. Dpy worms are short, fat, and often slow-moving or sluggish. Dumpy phenotype worms have reduced fertility and/or viability, making it impossible to maintain the population of homozygotes necessary for the experiments. Therefore, we have put a recessive lethal mutation on the sC1 balancer via CRISPR. This eliminates these sC1 homozygotes that are unusable worms and hugely increases the % of worms usable for analysis in the culture. To correct this significant dpy issue, we proceeded to genetically engineer the sC1 balancer via CRISPR, deleting the let-805 gene copy that is on the balancer. Heterozygous worms to the balancer are wild type (since the gene is haplosufficient), allowing the propagation of the strain; homozygous fem-2 mutants are wild type and are homozygote wild type for let-805. The only homozygous worms for the let-805 deletion are also homozygous to the balancer that carry the dpy-1(s2170) allele. This edit almost completely eliminated the dpy worms and increased the percentage of worms usable for analysis in culture. Our preliminary analysis indicates that the current percent of dyp in the population

Task Progress:

is below 20% and most of these worms die as adults and therefore do not reproduce. We plan to edit a deletion of LEA-1 into this strain to provide the suitable control (in the same genetic background). These genetically engineered strains allow study of the accumulation of double stranded DNA breaks without repair, and control of life cycle at a single point, preventing complications of multiple life cycle eggs, larvae, and adult worms in the analysis, while maintaining fertility and viability of a heterozygous population for study.

Specific Aim 2A.

In specific aim 2A, for LEA-1 deletion, we assessed if LEA-1 recruitment to double stranded breaks (DSBs) occurs after micro irradiation, validating the dynamic range and sensitivity of the assays for the cosmic radiation simulation studies proposed. Since we are working with a new microscope (Nikon SoRa) we had to test the micro irradiation conditions on known (published) protein (MRE-11), which has been successfully completed. We now have everything set with the methodology. However, since LEA-1 is cytoplasmic, we have crossed the LEA-1 reporter strain to a histone marked strain (so we can optimally aim the laser accurately). LEA-1 mutants are homozygous viable, with no effect on oocytes, allowing us to assay chromosome morphology and number.

Specific Aim 1: B) Identify and quantification of DNA damage by Surface Enhanced Raman Spectroscopy (SERS) and Specific Aim 2: To test whether the intrinsically disordered protein molecular shield, late embryogenesis abundant protein (LEA) protects from cosmic radiation damage

Bottom line up front (BLUF):

These protocols investigated how UV radiation affects the molecular integrity of C. elegans, specifically comparing wild-type worms to LEA-1 mutants, which lack radiation protective proteins. The analysis uses the advanced imaging techniques Raman spectroscopy and Surface Enhanced Raman Scattering (SERS) imaging, high-speed, label-free, and 3D microscopy techniques that create chemical maps of living cells and tissues by detecting intrinsic molecular vibrations. We identified significant DNA damage and biomolecular stress following UV exposure, noting that mutants were more vulnerable to degradation. By optimizing chemical tagging with gold nanostars and correlating fluorescence with Raman data, we are developing a high-resolution toolkit to map how radiation disrupts lipids, proteins, and DNA across different life stages. This is the first use of these techniques on whole living organisms for spaceflight related purposes preparatory for assaying effects of simulated cosmic radiation.

Specific details:

We have been investigating the impact of UV radiation exposure on genomic integrity and germline dynamics in C. elegans, using wild type (N2), LEA-1 mutants (LP847), and a few positive control strains. The experimental workflow included synchronized larval staging, controlled UV irradiation at defined doses, and high-resolution DAPI imaging to evaluate germline structure and nuclear condensation. DAPI (4′,6-diamidino-2-phenylindole) is a widely used fluorescent stain for labeling double-stranded DNA (dsDNA) in fixed cells.

In parallel, we performed extensive optimization, including substrate tuning and laser wavelength/intensity calibration, to achieve reproducible spontaneous Raman and SERS spectra from Carnoy-fixed worms. We also developed and validated an in vivo tagging approach using gold nanostars for molecular analysis via SERS. The gold nanostars provide a surface that increases analytical sensitivity orders of magnitude, while limiting background interference. In addition, we explored Raman imaging within specific regions of interest in C. elegans to spatially map proteins (amide I band), lipids, and nucleic acids (DNA) using both spontaneous Raman and SERS. Fluorescently tagged AuNPs were also employed to label C elegans, enabling colocalization through fluorescence microscopy.

We are also working on an integrated correlative Raman and widefield fluorescence imaging technique to achieve dual-mode mechanistic insights into radiation-induced molecular alterations in C. elegans.

As an initial feasibility study, Raman spectroscopy was employed to evaluate biomolecular damage in C. elegans following UV radiation exposure (200 J). All experiments were conducted on L4-stage worms. Carnoy-fixed N2 wild-type worms and LP847 (lea-1 null mutant) worms were analyzed under control and UV-treated conditions, with approximately ten spectra collected per group. To increase signal-to-noise, three consecutive acquisitions are integrated into each spectrum.

In both strains, comparative spectral analysis showed distinct and consistent differences between UV-exposed and control worms. After UV exposure, Raman vibrations corresponding to DNA backbone integrity (phosphate stretching) showed decreased intensity and increased peak broadening, which is attributed to radiation-induced modifications in the phosphodiester backbone. In comparison to N2, these UV-induced alterations were consistently more noticeable in LP847 worms, suggesting more DNA damage in the absence of LEA proteins. In addition to DNA-specific signatures, UV exposure induced broader spectral changes consistent with global biomolecular stress. Ongoing analysis is focused on resolving additional biomolecular modifications reflected in these spectral regions and establishing their relationship to radiation-induced damage pathways.

In order to supplement point-spectroscopy measurements and enable spatially resolved assessment of radiation effects, stimulated Raman scattering (SRS) imaging has been started to map the 2850 cm^{-1} Raman band associated with lipid CH_2 stretching across the intact/live worm. This method allows lipid distribution and possible radiation-induced lipid remodeling to be seen throughout the entire animal. Initial imaging experiments are underway, and analysis is ongoing to determine strain- and treatment-dependent differences. Stimulated Raman Scattering (SRS) utilized here uses two lasers to intensely stimulate vibrations for rapid imaging. SERS is ideal for trace detection, while SRS offers the faster acquisition and better quantification optimal for this study.

These initial findings show that Raman-based methods can accurately identify radiation-induced molecular damage in C. elegans in a variety of biomolecular classes. Future work will concentrate on quantitative analysis using peak intensity ratios and difference spectra, extending these studies to space-relevant radiation conditions, and evaluating developmental-stage-dependent responses by expanding analyses to embryonic (egg), early larval (L1/L2), and adult stages to directly evaluate LEA-mediated protection across the life cycle.

Design, construction, and validation of new flight hardware.

Bottom line up front (BLUF): The design and construction of the new flight hardware is complete and has passed key validation milestones. The system's pumps are performing with high precision, and the electronics have proven reliable during simulated mission cycles. Collaborating with Bionetics, we finalized assembly of a prototype, and both the

	ground and flight units are complete. Validation and calibration studies show that the hardware pumps all deliver within 2% volume of each other. The electronic system works perfectly during cycles of the mission profile. The engineering components of the ground and flight units include a battery pack for power, a control board with a 3D gravity sensor, temperature, humidity and power load sensors, and two reservoirs with motorized pumps to deliver fluids to culture bags. Each unit accommodates eight 10-mL Teflon bags with substantial O2 and CO2 permeability, but high water retention. Pumps from each reservoir connect the reservoirs to the bags with live nematodes, to refeed 2X a day, via a manifold and tubing, until it is time to add termination fluid at the completion of the experiments. The experiments are designed to not need toxic chemical fixatives. Validation and testing of all pumps, and components, continues. Interactions. Seamless and constructive interactions between all stakeholders are ongoing.
Bibliography Type:	Description: (Last Updated: 03/31/2026)
Abstracts for Journals and Proceedings	Hammond T, Smolikove S, Panikkanvalappil S, Birdsall H. "Protection from cosmic radiation: An Artemis II based initiative." 41st Annual Meeting of the American Society for Gravitational and Space Research, Phoenix, Arizona, December 3-6, 2025. Abstracts. 41st Annual Meeting of the American Society for Gravitational and Space Research, Phoenix, Arizona, December 3-6, 2025. , Dec-2025
Articles in Peer-reviewed Journals	Hammond TG, Panikkanvalappil SR, Allen PL, Gaikani HK, Nislow C, Giaever G, Zhang Y, Levine HG, Gaza R, Dimapilis D, Wells HW, Russick JM, Durand PM, Birdsall HH. "Programmed cell death and redox metabolism protect Chlamydomonas reinhardtii populations from the galactic cosmic environment on the Artemis-1 mission." Sci Rep. 2025 Jul 2;15:23396. https://doi.org/10.1038/s41598-025-05419-w ; PMID: 40603369; PMCID: PMC12222459 , Jul-2025